

Last updated on September 7, 2026

Activities

[2024–] **Senior Principal Research Scientist** at Mitsubishi Electric Research Laboratories (MERL)

[2022–2024] **Principal Research Scientist** at Mitsubishi Electric Research Laboratories (MERL)

[2019–2022] **Assistant Research Scientist** (*Inv. Aux.*) at Instituto Superior Técnico, Lisbon

[2018–2019] **Postdoctoral Associate** at KTH Royal Institute of Technology, Stockholm

[2014–2018] **Postdoctoral Researcher** at Instituto Superior Técnico, Lisbon

Education

[2003–2007] **BSc degree** at Electrical and Computer Engineering, University of Coimbra.

[2007–2008] **MSc degree** at Electrical and Computer Engineering, University of Coimbra.

[2009–2013] **PhD degree** at Electrical Engineering, University of Coimbra.

Research

Selected Recent Papers

[T10] “Revisiting Monocular SLAM with Spatio-Temporal Scene Modeling” (with Piedade, Manam, and Yamazaki). In: *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. [[code](#)]. 2026.

[T9] “Point4Cast: Streaming Dynamic Scene Reconstruction and Forecasting” (with Liu, Lohit, Jiang, Sawada, Tai, Tang, and Chatterjee). In: *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. **Highlight paper**, acceptance rate of about 3.6%. 2026.

[T8] “LASER: Layer-wise Scale Alignment for Training-Free Streaming 4D Reconstruction” (with Ding, Xie, Liang, Chatterjee, and Jiang). In: *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. [[arXiv](#),[code](#)]. 2026.

[T7] “SAC-GNC: Sample Consensus for adaptive Graduated Non-Convexity” (with Piedade, Sidhartha, Gaspar, and Govindu). In: *IEEE/CVF Int’l Conf. Computer Vision (ICCV)*. [[code](#),[doi](#)] **Highlight paper**, acceptance rate of about 2.3%. 2025.

[T6] “A Probability-guided Sampler for Neural Implicit Surface Rendering” (with Pais, Piedade, Chatterjee, and Greiff). In: *European Conf. Computer Vision (ECCV)*. [[doi](#)]. 2024.

[T5] “Gear-NeRF: Free-Viewpoint Rendering and Tracking with Motion-aware Spatio-Temporal Sampling” (with Liu, Tai, Tang, Lohit, and Chatterjee). In: *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. [[code](#),[doi](#)] **Highlight paper**, acceptance rate of about 2.8%. 2024.

[T4] “BANSAC: A dynamic BAYesian Network for adaptative SAmple Consensus” (with Piedade). In: *IEEE/CVF Int’l Conf. Computer Vision (ICCV)*. [[arXiv](#),[code](#),[doi](#)]. 2023.

- [T3] “Robust Frame-to-Frame Camera Rotation Estimation in Crowded Scenes” (with Delattre, Dirnfeld, Nguyen, Scarano, Jones, and Learned-Miller). In: *IEEE/CVF Int’l Conf. Computer Vision (ICCV)*. [[arXiv](#),[code](#),[doi](#)]. 2023.
- [T2] “Fast and Accurate 3D Registration from Line Intersections Constraints” (with Mateus, Ranade, and Ramalingam). In: *Int’l J. Computer Vision (IJCV)* 131 (2023). [[code](#),[doi](#)], pp. 2044–2069.
- [T1] “A Unified Model for Line Projections in Catadioptric Cameras with Rotationally Symmetric Mirrors” (with Iglesias). In: *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*. [[doi](#)]. 2022, pp. 15776–15785.

Patents

- [P8] “Neural Radiance Field Training Based on a Non-Uniform Sample of Image Pixels” (with Pais, Greiff, and Chatterjee). US20250272909A1 & WO2025177708A1. **Granted**. May 2026. URL: <https://patents.google.com/patent/US20250272909A1>.
- [P7] “Oriented-grid Encoder for 3D Implicit Representations” (with Pais and Gaur). US20250131651A1 & JP2025071766A. **Granted**. May 2026. URL: <https://patents.google.com/patent/US20250131651A1>.
- [P6] “System and Method for Monocular Localization and Mapping Using Temporal and Spatial Support Retrieval” (with Piedade and Manam). **Filed**. May 2026.
- [P5] “System and Method for Joint Vehicle Positioning and Map Estimation using a Compound Probabilistic Filter” (with Berntorp, Greiff, and Di Cairano). US20260072434A1 & JP2024137659A. **Allowed**. Mar. 2026. URL: <https://patents.google.com/patent/US20260072434A1>.
- [P4] “Neural radiance field training using spatially dynamic loss functions” (with Pais and Chatterjee). US20260057658A1 & WO2026042893A1. **Published**. Feb. 2026. URL: <https://patents.google.com/patent/US20260057658A1>.
- [P3] “Rendering Two-Dimensional Image of a Dynamic Three-Dimensional Scene” (with Chatterjee and Lohit). US12475636B2 & WO2025154553A1. **Granted**. Nov. 2025. URL: <https://patents.google.com/patent/US12475636B2>.
- [P2] “Dynamic (4D) Scene Reconstruction Using Multiple Neural Radiance Fields” (with Liu, Chatterjee, and Lohit). US20250308153A1 & WO2025206410A1 & EP4670128A1. **Allowed**. Oct. 2025. URL: <https://patents.google.com/patent/US20250308153A1>.
- [P1] “Enhanced Computation of Robust Loss Functions using Adaptive Annealing Factors” (with Piedade). **Filed**. Mar. 2025.

Teaching and Supervision

Courses

Course #5: Artificial Intelligence and Decision Systems (2021–2022).

Course #4: Artificial Intelligence and Decision Systems (2020–2021).

Course #3: Computer Graphics (2017–2018).

Course #2: Algorithms & Data Structures (2016–2017).

Course #1: Automatic Control (2015–2016).

Students

Doctor of Philosophy (PhD) Students

Student #2: Gonalo Dias Pais (2017–2021):

Localize and Map Dynamic Environments with 3D Sensors and Semantic Information [[link](#)].

Student #1: Andr  Mateus (2017–2021):

On the Exploitation of 3D Straight Lines for Active Mapping and Camera Localization [[link](#)].

Ongoing PhD Students

PhD Student #1: Valter Piedade (2022–2026):

Real-time Visual-SLAM in Indoor Environments.

Interns

Intern #10: Iason Velentzas (2026) at MERL, from Georgia Institute of Technology, USA.

Intern #9: Dave Dirnfeld (2026–2027) at MERL, from UMass Amherst, USA.

Intern #8: Valter Piedade (2025) at MERL, from Instituto Superior T cnico, Lisbon, Portugal.

Intern #7: Aniket Gupta (2025) at MERL, from Northeastern University, USA.

Intern #6: Prajnan Goswami (2025) at MERL, from Northeastern University, USA.

Intern #5: Fabien Delattre (2023) at MERL, from UMass Amherst, USA.

Intern #4: Jiachen Liu (2023) at MERL, from Penn State University, USA.

Intern #3: G. Dias Pais (2022) at MERL, from Instituto Superior T cnico, Lisbon, Portugal.

Intern #2: Arihant Gaur (2022) at MERL, from Visvesvaraya National Institute of Technology, Nagpur, India.

Intern #1: Arihant Gaur (2021–2022) at IST, from Visvesvaraya National Institute of Technology, Nagpur, India.